

HIGHLAND THREADS

11700 GLOGER, HOUSTON, TX 77039
Phone: 800-847-5713 / 281-986-5100 / Fax: 281-986-5151



Material Test Report

MTR #: 1322906

Page: 1 of 1

SOLD TO: BIG RED FASTENERS
P.O. BOX 470547
TULSA, OK 74147

SHIP TO: BIG RED FASTENERS
489 NORTH POPLAR AVENUE
RECEIVING
BROKEN ARROW, OK 74012

DATE	SALES ORDER #	CUST P.O.#	TAG NUMBER	SPECIFICATION	
01/05/2026	1748122	B-00059140		ASTM A193 B7, ASTM A320 L7	
LINE #	QTY	ITEM #	ITEM DESCRIPTION	LOT NUMBER	MILL HEAT
2	40		1 3/8-8 X 11 1/4 ASTM A320 L7 ALL THREAD STUDS Xylan 1424 BLUE COATED- USA	CUKG	224016

CHEMICAL ANALYSIS						
C	Mn	P	S	Si	Cr	Mo
.42	.88	.013	.022	.23	.833	.159

PHYSICAL PROPERTIES						
Elongation %	Reduction of Area %	Tensile PSI	Yield PSI	Hardness Brinell	Charpy @ -150 F FtLbF	Charpy @ -150 F FtLbF MIN. TEMPERING:
21.42	60.47	139545	125395	301	24	21 1100 F.

ASTM E381: S2/R1/C2 ASTM A193-20 ASTM A320-18 QUENCHED AND TEMPERED
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE
REQUIREMENTS OF THE REFERENCED SPECIFICATION
COUNTRY OF ORIGIN USA

Stacy Kallers
Quality Assurance Department

Job: 74769
PO: 7476910
BOM: 36
Pages: 3
Heat: 224016
62133857
A76790



Material Test Report

MTR #: 1322907

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DATE		SALES ORDER #		CUST P.O.#		TAG NUMBER		SPECIFICATION	
01/05/2026		1748122		B-00059140				ASTM A194 GRADE 7/7L HVY HEX NUTS 1 3/8-8	
LINE #	QTY	ITEM #	ITEM DESCRIPTION			LOT NUMBER		MILL HEAT	
3	80		1 3/8-8 ASTM A194 GRADE 7L HEAVY HEX NUTS Xylan 1424 BLUE COATED-USA			CFPV		62133857	

CHEMICAL ANALYSIS						
C	Mn	P	S	Si	Cr	Mo
.42	.84	.010	.020	.20	.93	.171

PHYSICAL PROPERTIES					
Hardness	Proof Load	24 HR Hard	Charpy @	Charpy @	MIN.
Brinell	Lbf	BHN	-150 F F _{LLbF}	-150 F F _{LLbF}	TEMPERING:
269	N/A	255	24	23	1100 F.

SUPPLEMENTAL INFORMATION					
ASTM E381: S1/R1/C1 ASTM A194-16 QUENCHED AND TEMPERED					
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE					
REQUIREMENTS OF THE REFERENCED SPECIFICATION					
***COUNTRY OF ORIGIN USA---					

Stacy Kellars
Quality Assurance Department



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MTR #: 1322908

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DATE		SALES ORDER #		CUST P.O.#		TAG NUMBER		SPECIFICATION	
01/05/2026		1748122		B-00059140				ASTM F436 HARDENED FLAT WASHERS	
LINE #	QTY	ITEM #	ITEM DESCRIPTION			LOT NUMBER		MILL HEAT	
4	80		1 3/8 F436 FLATWASHERS XYLAN 1424 BLUE COATED- USA			CLVW		A76790	

CHEMICAL ANALYSIS

C	Mn	P	S	Si	Cr	Ni
.52	.72	.011	.001	.20	.09	.04

PHYSICAL PROPERTIES

Hardness RC
43.1

SUPPLEMENTAL INFORMATION

ASTM F436-11
MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE REQUIREMENTS OF THE REFERENCED SPECIFICATION
COUNTRY OF ORIGIN USA

Steve Fellers
Quality Assurance Department